

Total No. of Questions : 8]

SEAT No. :

PD4639

[Total No. of Pages : 2

[6404]-145

B.E.(E & TC) (Electronics)

REMOTE SENSING

(2019 Pattern) (Semester - VIII) (404192 C) (Elective-VI)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1) a)** Write a short note on **[8]**
- i) False color composite
 - ii) True color composite
- b)** Enlist the difference between raster data and vector data. **[8]**

OR

- Q2) a)** Explain various band combinations and color composites used in satellite image processing with suitable examples of LANDSAT 8 bands. **[8]**
- b)** Compare supervised and unsupervised classification. **[8]**
- Q3) a)** Differentiate between active and passive microwave remote sensing. **[8]**
- b)** Explain concept of SAR and SLAR with suitable diagram & Brief about the characteristics and interpretation of SAR & SLAR data. **[10]**

OR

- Q4) a)** Explain LIOAR concept, its types, characteristics with applications. **[10]**
- b)** Explain RADAR equation and determine ground based resolution and azimuth resolution of SLAR with suitable illustrations. **[8]**
- Q5) a)** Explain GNSS technology and signal structure with suitable diagram. **[8]**
- b)** Explain various segments of GNSS technology. **[10]**

OR

P.T.O.

- Q6)** a) Explain space based augmentation system GAGAN. [10]
b) Explain different tools for GNSS data processing. [8]

- Q7)** a) Brief the application of remote sensing in coastal side with use cases and methodology. [10]
b) Brief the application of remote sensing in agriculture with use cases and methodology. [8]

OR

- Q8)** a) Brief the application of remote sensing in urban sprawl planning and management with use cases and methodology. [10]
b) Brief the application of remote sensing in water resource management with use cases and methodology. [8]

